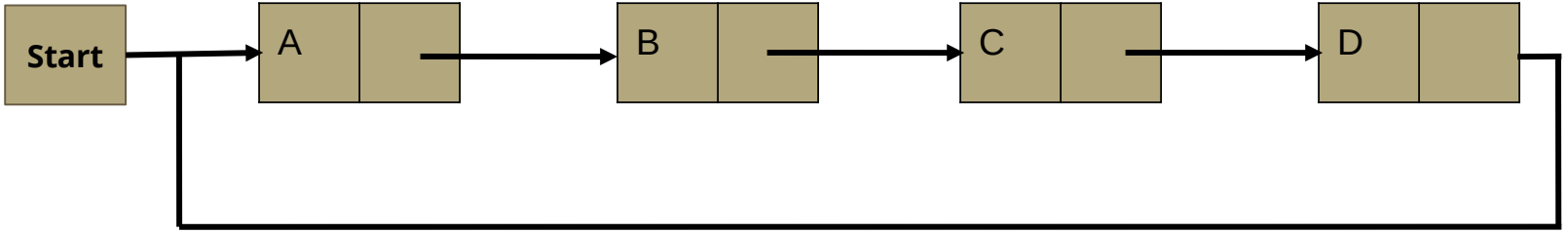


Data Structure(CSL202)

By-
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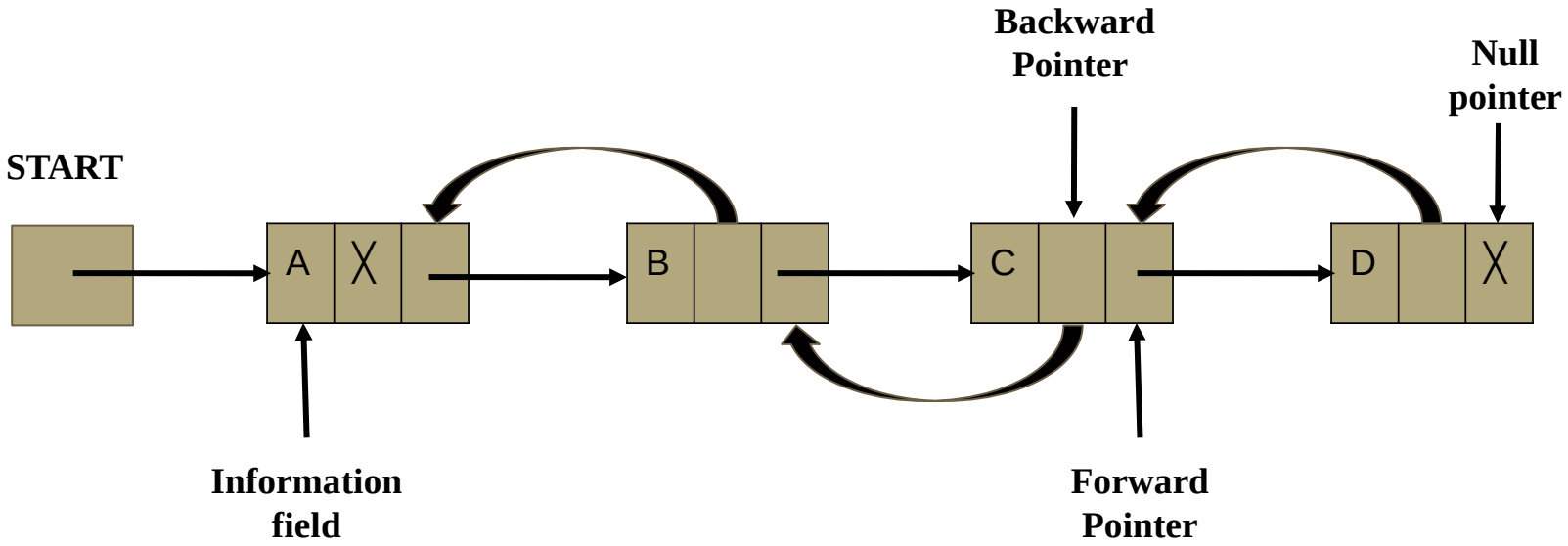
Circular Linked List

- Last node points back to the first node instead of containing the null pointer, called a circular list.

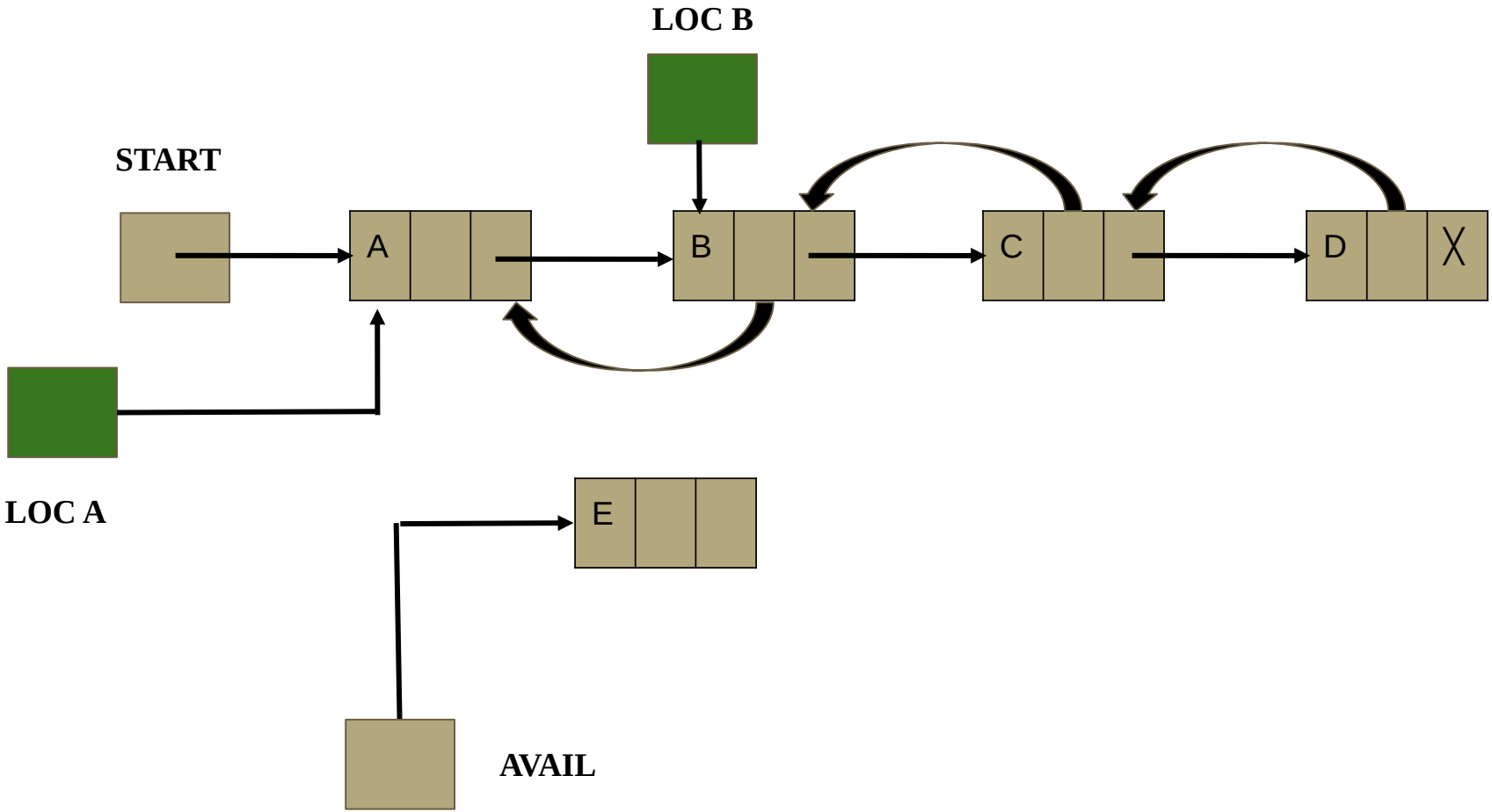


Doubly Linked List

- Made up of nodes where each node contains:
  - **INFO** – the actual data.
  - **Forward Pointer –** Address of the next node.
  - **Backward Pointer –** Address of the previous node.



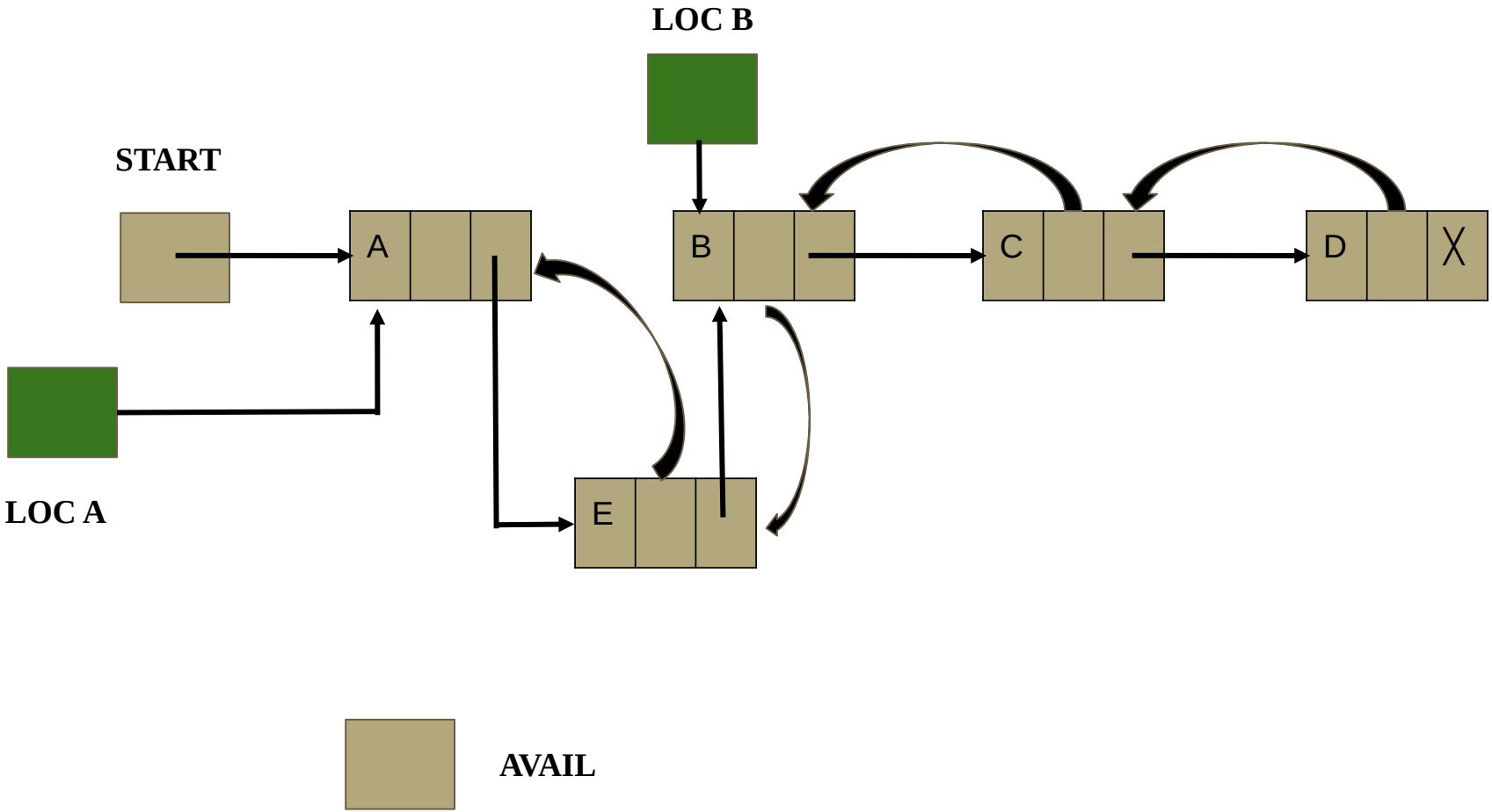
Insert Between Two Locations in Doubly Linked List



Insert Between Two Locations in Doubly Linked List

```
1. If AVAIL == NULL:
       // No new node
      Exit.
2. NEW := AVAIL
   AVAIL := FORW[AVAIL]                    // Remove a node from the AVAIL list
   INFO[NEW] := ITEM                       // Copy ITEM into the new node
3. FORW[LOCA] := NEW              // LOCA's next becomes NEW
   FORW[NEW] := LOCB                       // NEW's next becomes LOCB
   BACK[LOCB] := NEW                       // LOCB's previous becomes NEW
   BACK[NEW] := LOCA                       // NEW's previous becomes LOCA
```

Insert Between Two Locations in Doubly Linked List



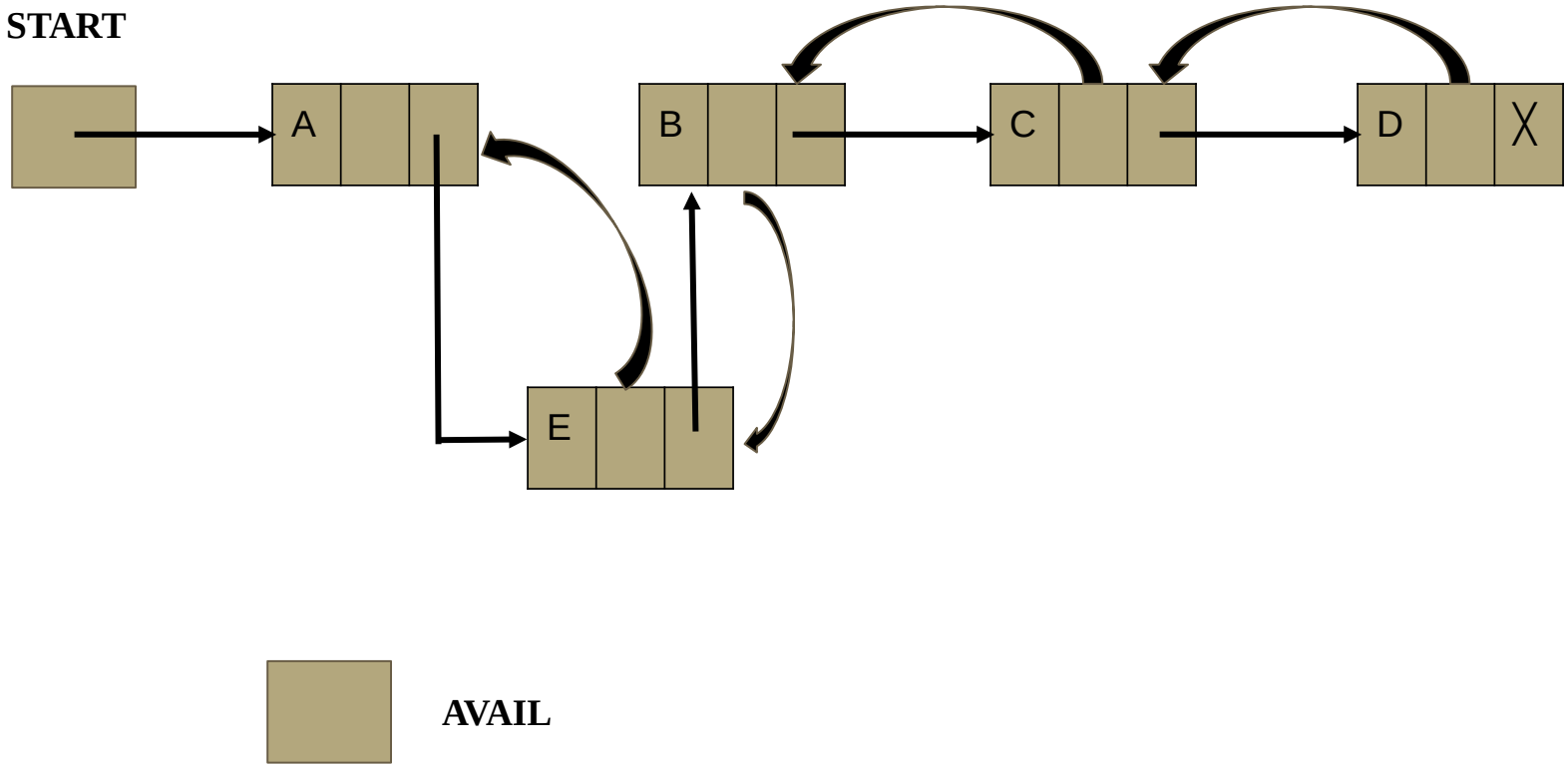
Insert Between Two Locations in Doubly Linked List

Time Complexity :

- Best Case: Θ(1)
- Average Case: Θ(1)
- Worst Case: Θ(1)

The algorithm performs constant-time pointer assignments.

Deleting a node from a Doubly Linked Lis.



Deleting a node from a Doubly Linked List

1. [Adjust the links of neighboring nodes to bypass LOC]

    FORW[BACK[LOC]] := FORW[LOC]

    BACK[FORW[LOC]] := BACK[LOC]

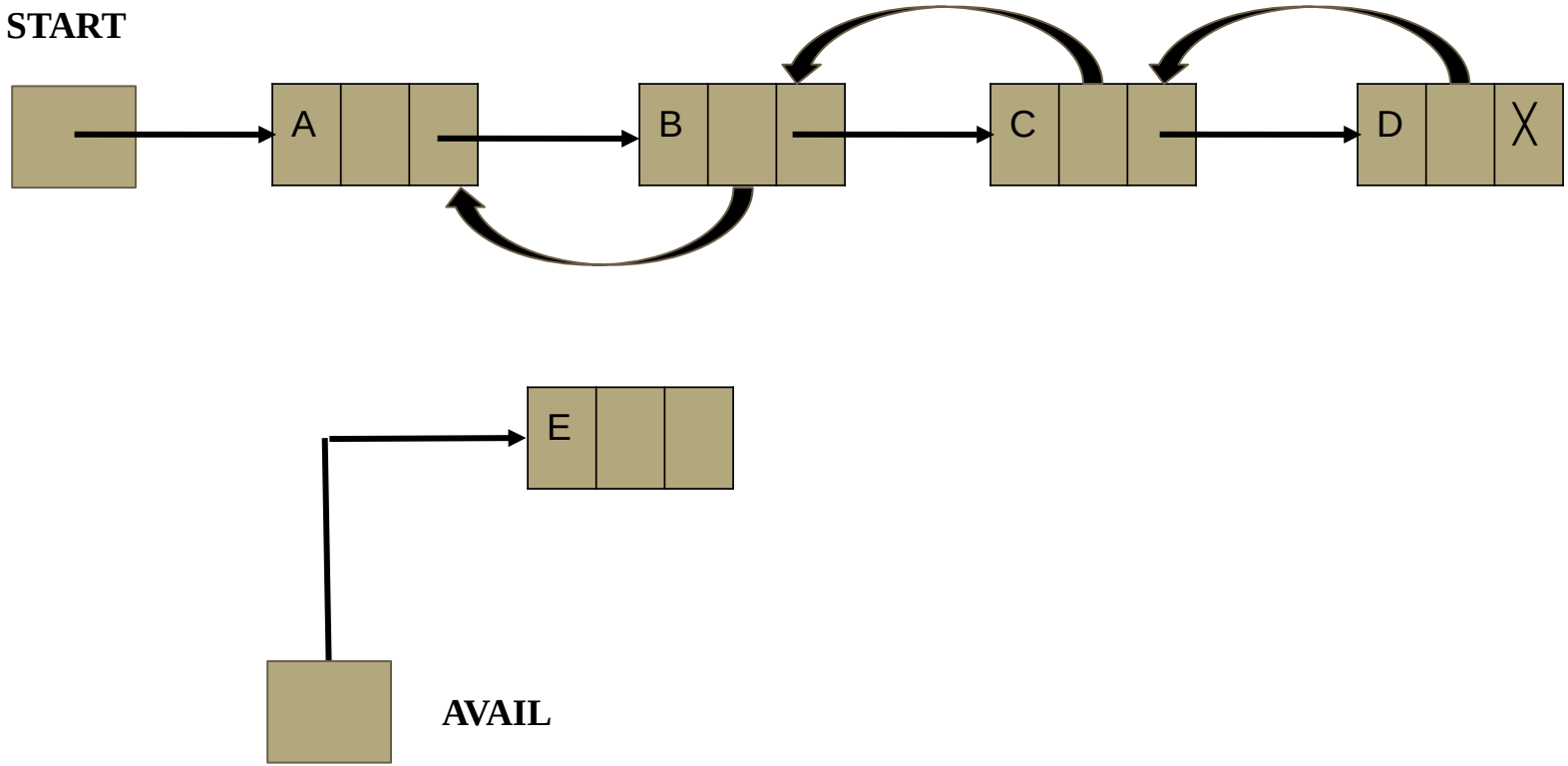
2. [Return deleted node to the AVAIL list]

    FORW[LOC] := AVAIL

    AVAIL := LOC

3. Exit.

Deleting a node from a Doubly Linked Lis



Deleting a node from a Doubly Linked List

Time Complexity :

- Best Case: Θ(1)
- Average Case: Θ(1)
- Worst Case: Θ(1)

The algorithm performs constant-time pointer assignments.